An Immersive 360 Video Streaming Dataset with Synchronized QoE and 5G Radio Metrics

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***Abstract*—Compared to traditional 2D videos, there has been an ever-growing demand for 360° videos in recent years. Therefore, Quality of Experience (QoE) is becoming extremely challenging for immersive videos in mobile networks due to varying network conditions. Despite tremendous improvements offered by 5G networks, mapping Quality of Service (QoS) metrics to the objective QoE for 360-degree videos remains of great interest to stakeholders and content providers. This mapping helps them understand the impact of network conditions and supports scalable testing in realistic scenarios. In this work, we provide a realistic 5G dataset while playing 360° video sessions on a Meta Quest 3 over a 5G network. Our dataset provides both radio metrics and QoE logs for YouTube at 1-second granularity. Moreover, we preprocessed the dataset by synchronizing QoE logs and 5G radio metrics using their associated timestamps. Therefore, for each streaming session, there are corresponding radio metrics, including RSRP, RSRQ, and SNR, as well as those from neighboring cells. The resulting datasets can be used by researchers to optimize 360° video streaming applications by investigating the impact of radio metrics. We believe that our dataset will enable further research in this exciting new direction.**

***Index Terms*—360° Videos, 5G, QoE, Immersive videos, dataset**

I. Introduction

The rapid growth of video streaming has made Quality of Experience (QoE) a central focus of wireless networks, particularly in the era of 5G. Among the many immersive media technologies gaining popularity, 360° video stands out as one requiring higher network stability and application-level responsiveness than traditional 2D video, since fluctuations in throughput and radio quality directly affect the viewer's experience. OTT platforms dynamically adjust video bitrate based on network conditions, making QoE highly sensitive to such fluctuations [1]–[12].

Understanding this sensitivity is particularly important for 360° content, where a degraded viewing experience is far more disruptive than in conventional video. A momentary drop in resolution or a buffering stall in a 2D clip is an inconvenience; the same event in an immersive headset can break presence, induce discomfort, or trigger motion sickness, since the visual field no longer matches the user's expectation of a continuous environment. As 5G networks become the default connectivity option for standalone headsets such as the Meta Quest, operators and content providers need a clear picture of how serving- and neighboring-cell radio conditions – RSRP, RSRQ, SNR – propagate into application-level behavior such as bitrate selection, quality switching, and buffering. Without datasets that capture both layers together at fine granularity, this mapping remains largely theoretical, limiting the development of radio-aware adaptation strategies and realistic QoE prediction models for immersive media.

In the literature, several works have studied user behavior and perception during 360° video viewing. One line of work recorded head movements of 59 users watching 5 70-second 360° videos on a Razer OSVR HDK2 headset, providing an empirical overview of viewing behavior in viewport-adaptive streaming [13]. This was extended by capturing both head and eye movements from 57 participants across 19 videos in a free-viewing experiment, additionally examining how the starting orientation of a video affects viewing patterns [16]. Another study recorded head rotations from 48 participants watching 20 omnidirectional videos on an HTC Vive, supplementing this with Simulator Sickness Questionnaire (SSQ) responses collected after each session [19]. A different approach paired content-derived data, such as image saliency and motion maps, with sensor data including viewer head positions and orientations from HMDs [14]. Head-movement traces from 48 participants watching 18 videos across 5 categories revealed that viewing patterns in VR spherical streaming differ noticeably from those in conventional video [15].

Similarly, a database of 73 immersive VR clips annotated with arousal and valence ratings alongside head-movement recordings uncovered a relationship between movement and perceived emotion [20]. Head and gaze data combined with a technical classification scheme for 360° content showed that head movements tend to follow gaze direction [17]. A focused eye-tracking study manually annotated gaze data from 13 observers and proposed a rule-based eye-movement classifier [18]. More recently, a large-scale labeled dataset of over 10k segmentation and tracking experiments spanning diverse indoor and outdoor scenes was introduced, partly constructed with the help of large language models during labeling, followed by manual revision [22]. A broader taxonomy for organizing 360° datasets, including viewer feedback and viewport-pattern analysis, was also proposed rather than collecting new measurements [21].

What this prior work shares is a focus on behavioral and perceptual aspects of 360° viewing – head movement, gaze, eye movement, viewport exploration, or affective responses [13]–[22] – largely decoupled from the underlying network. None

TABLE I
COMPARISON OF OUR DATASET WITH EXISTING 360° VIDEO DATASETS

Category	Works	Head Movement	Gaze	Subjective QoE	QoE at 1s	5G Radio at 1s	YouTube QoE Logs
Head-movement datasets	[13]–[15]	Yes	No	No	No	No	No
Head/gaze datasets	[16], [17]	Yes	Yes	No	No	No	No
Gaze-only / SSQ datasets	[18], [19]	Partial	Yes	Partial	No	No	No
Affective-rating datasets	[20]	Yes	No	Yes	No	No	No
Taxonomy / large-scale	[21], [22]	No	No	No	No	No	No
This work	—	No	No	Yes	Yes	Yes	Yes

of these datasets jointly capture application-layer QoE and radio-layer measurements at fine temporal granularity, leaving a gap for researchers interested in how cellular network conditions translate into observable QoE degradation during real immersive playback. Our study is therefore not centered on behavioral saliency or viewing-pattern modeling alone; instead, it is designed as a cross-layer dataset to understand how wireless conditions affect QoE during immersive 360° video streaming.

Our work addresses this gap directly. We collect 360° video QoE every second while users watch YouTube 360° content on a Meta Quest 3 device connected through a real 5G hotspot, and in parallel, we record per-second 5G radio metrics from the smartphone, including RSRP, RSRQ, SNR, and neighboring-cell information. As of now, no existing dataset jointly combines immersive 360° video experience, continuous per-second QoE logging, and real 5G radio measurements in a unified experimental setting, as summarized in Table I.

Concretely, our contribution is threefold. *(i)* we release a synchronized dataset of 34 controlled streaming sessions covering 14 distinct 360° videos across categories such as travel, wildlife, sports, and documentary content, each paired with per-second serving- and neighbor-cell radio measurements collected via G-NetTrack. *(ii)* we provide an initial characterization of the dataset, showing how radio quality varies across experiments and how this variation relates to download bitrate and resolution distribution. *(iii)* we outline a set of concrete use cases – including QoE prediction, radio-aware bitrate adaptation, and LLM-based network-to-text summarization – that illustrate how the dataset can be applied in practice. Together, these contributions aim to give researchers a realistic, ready-to-use resource for studying the radio-to-QoE relationship in immersive streaming, an area where public data has so far been notably absent. The synchronized QoE and 5G radio dataset is available on GitHub.[1]

II. METHODOLOGY

The dataset collection pipeline consists of the following components: i) Meta Quest 3 – a Virtual Reality Headset Without Wires [2], ii) Samsung Galaxy S25+ operating as a mobile hotspot to connect with Meta Quest 3, and iii) YouTube iframe API to stream 360 videos – an application made to collect QoE logs every 1-second. The experimental setup and collection pipeline are shown in Figure 1. We selected publicly

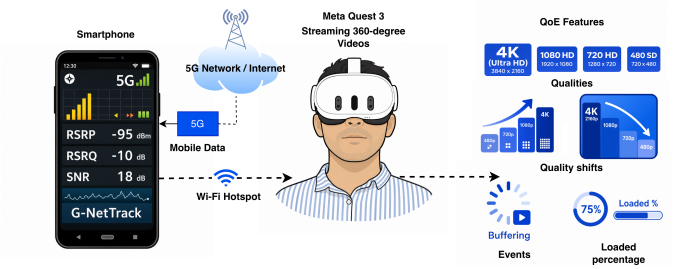


Fig. 1. Experimental setup and cross-layer data collection pipeline for 360° video streaming over 5G.

available videos on YouTube across categories such as Travel, Animation, Wildlife, and trailers. Moreover, videos are not too long, and we consider an average length of 4 minutes.[3]

In this setup, S25+ acted as the network access point providing the 5G radio connection for streaming. This setup enabled us to capture both application-layer QoE behavior and wireless radio conditions during the same streaming session. On the application side, the custom streaming application saves the following information to the MySQL: i) current resolution, ii) video bytes downloaded, iii) timestamp (which is later used to synchronize QoE and radio metrics), and QoE events, such as buffering, stop, pause, streaming, and ended. These events capture temporal changes in the streaming session and observe how playback behavior evolves under varying network conditions.

On the network side, the smartphone records 1-second radio measurements using the G-NetTrack application, including key indicators such as RSRP, RSRQ, and SNR of the serving and neighbor cells. All experiment logs are then synchronized to form a time-aligned dataset suitable for detailed statistical analysis. We show an example of a 5-second synchronized dataset in Table II. In the Table, the first two columns have the date and time, followed by the YouTube video ID titled as " London, United Kingdom. Virtual travel. 360 video in 8K " with 6:12/minutes of duration. We show 5 seconds of sample stats in the table, and during this time, the quality remains HD1080, as shown in the Quality column. Next, loaded % shows the current buffer, followed by 5G radio metrics, which are RSRP, RSRQ, and SNR, in the next three columns, along with download and upload bitrate. The next columns present the second cell RSRP, RSRQ, and SNR values.

[1]https://github.com/razaulmustafa852/meta-360-data

[2]Thirty Percent Sharper Resolution and 2X Graphical Processing

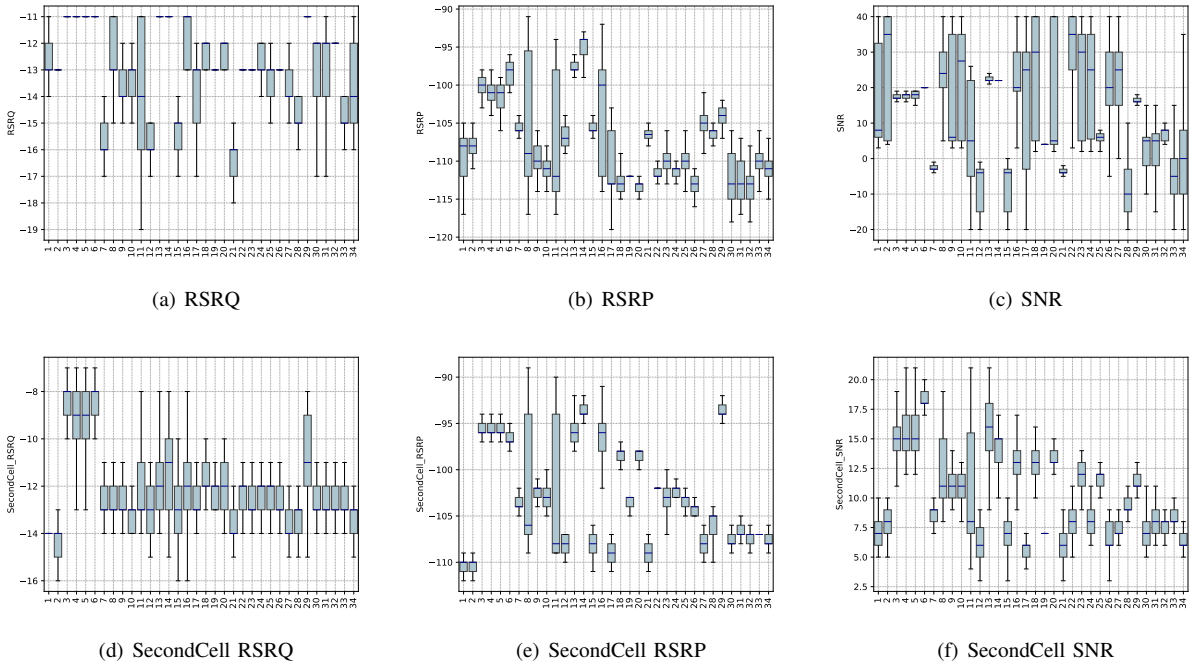
[3]The videos used in the experiments are available in the GitHub repository: https://github.com/razaulmustafa852/meta-360-data

TABLE II
SYNCHRONIZED QOE AND 5G RADIO-METRIC RECORDS.

Date	Time	Video	Quality	Loaded%	RSRP	RSRQ	SNR	DL_bitrate	UL_bitrate	SecondCell_RSRP	SecondCell_RSRQ	SecondCell_SNR
2025-10-31	11:49:30	KGerjHMa90s	hd1080	2.3	-111.0	-13.0	35.0	3.0	3.0	-108.0	-13.0	10.0
2025-10-31	11:49:31	KGerjHMa90s	hd1080	3.1	-111.0	-12.0	40.0	22173.0	7193.0	-107.0	-13.0	10.0
2025-10-31	11:49:32	KGerjHMa90s	hd1080	3.1	-111.0	-12.0	40.0	1.0	1.0	-107.0	-13.0	10.0
2025-10-31	11:49:33	KGerjHMa90s	hd1080	4.1	-112.0	-13.0	35.0	0.0	0.0	-107.0	-13.0	10.0
2025-10-31	11:49:34	KGerjHMa90s	hd1080	5.5	-112.0	-13.0	35.0	2.0	2.0	-107.0	-13.0	10.0

TABLE III
PERCENTAGE DISTRIBUTION OF STREAMING RESOLUTIONS ACROSS 34 EXPERIMENTS. ROWS ARE ORDERED FROM HIGHER TO LOWER RESOLUTION. – DENOTES 0% VALUES IN THE DATA.

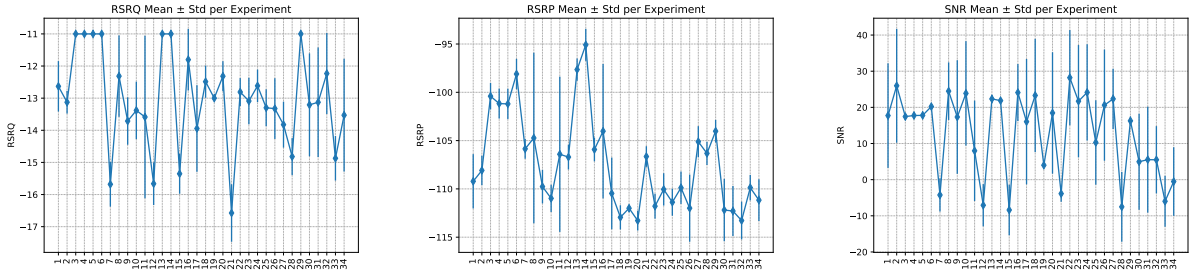
Quality	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
hd2160	21.15	–	100.0	100.0	100.0	100.0	100.0	80.39	28.17	3.12	57.72	100.0	100.0	100.0	100.0	100.0	96.09	100.0	97.61	100.0	100.0	100.0	4.17	100.0	–	100.0	–	96.4	98.21	96.1	59.91	92.34	100.0	100.0
hd1440	31.07	77.46	–	–	–	–	–	7.79	51.55	11.25	34.68	–	–	–	–	–	3.91	–	1.49	–	–	–	5.36	–	44.67	–	65.83	3.6	0.71	–	25.99	5.11	–	–
hd1080	46.74	22.54	–	–	–	–	–	11.82	20.28	85.62	7.6	–	–	–	–	–	–	–	0.9	–	–	–	90.48	–	55.33	–	28.84	–	1.07	3.9	14.1	2.55	–	–
hd720	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	5.33	–	–	–	–	–	–	–
large	1.04	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–



(a) RSRQ (b) RSRP (c) SNR

(d) SecondCell RSRQ (e) SecondCell RSRP (f) SecondCell SNR

Fig. 2. 5G Radio Metrics – RSRP, RSRQ, SNR for the Serving and Neighbor Cell.



(a) RSRQ (b) RSRP (c) SNR

Fig. 3. 5G Radio Metrics Mean/STD – RSRP, RSRQ, SNR for the Serving Cell.

III. DATASET STATISTICS AND CHARACTERIZATION

In this section, we present the dataset statistics and characterization for all 34 controlled experiments, with a focus on the distribution. Figure 2(a–c) presents the serving-cell RSRP, RSRQ, and SNR values, and Figure 2(d–f) shows the corresponding second-cell (neighbor) measurements. The mean and

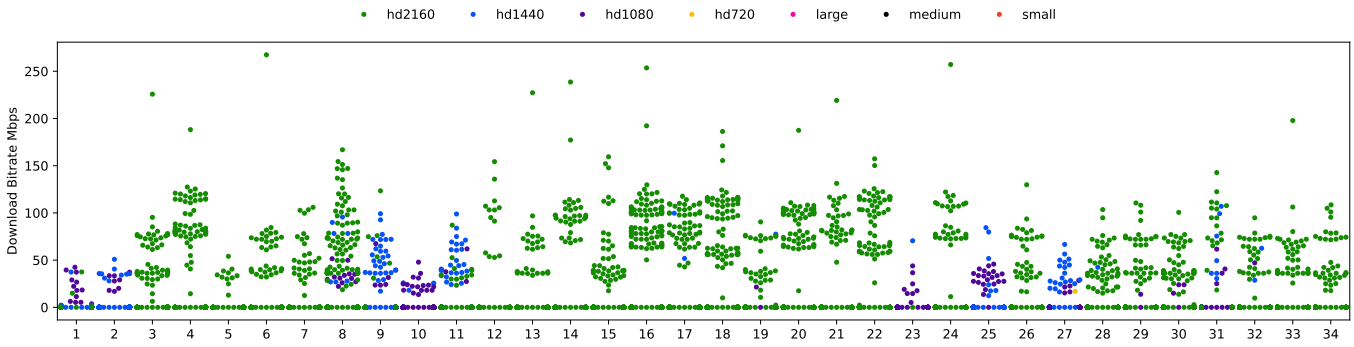


Fig. 4. Download Bitrate in Mbps across experiments.

standard deviation of serving-cell metrics are summarized in Figure 3. We also show the corresponding download bitrate in Figure 4 and quality distributions in Table III.

From figures 2 (a), we can see RSRQ ranges roughly from -11 dB to -17 dB, whereas 2 (b), RSRP varies from relatively strong cases around the mid -90 dBm region to weak cases near -115 dBm. Similarly, SNR values range from positive values above 20-30 dB to negative values. We also observe a dominant resolution of HD2160 across most experiments, with very few using HD1440 or HD1080, indicating that the system often remains above the minimum adaptation threshold even when channel quality degrades. Overall, RSRP spans relatively strong to weak received power levels, followed by RSRQ from less degraded to clearly impaired channel quality, and SNR from highly favorable positive values.

A few experiments, especially around 13-18 and 20-24, show the strongest overall performance. These experiments show favorable RSRP, RSRQ, and SNR, resulting in high download bitrate with higher resolutions, such as HD2160. These groups also exhibit large bitrate peaks, indicating that the user experiences high-resolution chunks that remain in the buffer, resulting in very few buffering events. This group of experiments represents the most stable end of the dataset, where strong 5G radio metrics conditions translate into high throughput and a reduced need for aggressive quality adaptation. In contrast, experiments such as 1-2, 7-12, 19, 23, 25, and parts of 26-34 show either weaker median 5G radio conditions or higher variability, and this is reflected in the QoE behavior.

We also show the neighbor cell RSRQ, RSRP, and SNR in Figure 2(d,e,f) across the 34 experiments. We observe that in some experiments, neighbor cell metrics show favorable conditions. For instance, we observe experiments 3-6 and 13-18, in which neighboring cells tend to have stronger RSRP, lower RSRQ degradation, and higher SNR. Therefore, our findings on the neighbor cells metrics is that good neighboring-cell conditions appear to indicate a healthier overall radio environment, in which the user is likely located in a strong coverage region with lower interference and better connectivity continuity. As a result, neighboring-cell metrics can be viewed as supportive indicators of stable high-throughput streaming, particularly when they align with good serving-cell conditions.

Please note that most streaming sessions are conducted at a 90° angle, meaning the user is watching the video content while sitting upright in a chair. RSRQ in experiments 7, 12, 16, 21, 28, and 33 experiences a more degraded average channel quality, therefore a larger uncertainty. Next, experiments at 3-6 and 13-16 show higher mean received power than at 18-20 and 30-34. SNR Compared to RSRQ, RSRP shows favorable conditions most of the time; while others drop toward near-zero or negative values and also show larger error bars, indicating substantial short-term fluctuation. Taken together, the dataset captures a wide range of 5G radio environments, from favorable to weak, which is essential for studying how changes in serving cell affect bitrate, quality switching, and buffering events in immersive video streaming.

IV. Application and Use Cases of the Dataset

The proposed dataset supports multiple research directions in immersive video streaming. Since the dataset synchronizes application-level QoE indicators with radio measurements from the serving and neighboring cells, it enables researchers to investigate how wireless network conditions affect 360° video quality, buffering behavior, bitrate adaptation, and playback stability. Table IV summarizes the main potential applications and research use cases of the dataset.

V. Conclusion

This paper presented a real-world dataset for analyzing the relationship between 5G radio metrics and Quality of Experience (QoE) in immersive 360° video streaming. The dataset was collected through 34 experiments conducted using a Meta Quest headset streaming YouTube 360° content over a 5G smartphone hotspot, while radio measurements were captured at fine time granularity using G-NetTrack. In addition to application-layer QoE features, the dataset includes both serving-cell and neighboring-cell radio metrics, enabling cross-layer analysis of how wireless network behavior affects adaptive video delivery. The collected traces span a wide range of radio environments, from stable and favorable sessions to weak and highly variable conditions, making the dataset suitable for studying bitrate dynamics, quality adaptation, and QoE degradation under realistic mobile network conditions. By releasing this dataset, we aim to support future research on QoE-aware streaming, radio-to-application performance

TABLE IV
POTENTIAL APPLICATIONS AND USE CASES OF THE 360° VIDEO STREAMING DATASET

Use Case	Description	Outcome
QoE Prediction	Joint radio and QoE logs allow ML models to forecast buffering events and quality switches (upgrades/degradations), either per timestep or at the session level.	Accurate QoE estimation models.
Radio-Aware ABR	1-second synchronization between radio and application metrics supports bitrate adaptation algorithms driven by signal quality.	Reduced quality fluctuations.
Serving vs. Neighbor Cell Impact	Correlations between serving/neighbor-cell radio metrics and QoE indicators such as buffering, switching, and initial delay.	Identification of key radio indicators for streaming quality.
ML Benchmarking	Time-synchronized data supports classification (QoE classes: Poor/Fair/Good/Excellent), regression (loaded % from RSRP/RSRQ/SNR), and time-series forecasting (e.g., LSTM).	Benchmark for ML methods in immersive streaming.
Edge/5G Optimization	Supports edge-assisted or network-aware delivery strategies for 360° video over 5G.	Improved resource allocation and delivery.
Synthetic Data Generation	Real traces can seed statistical or LLM-based synthetic data generation, given the difficulty of large-scale collection.	Scalable training data for QoE models.
LLM-based QoE Reasoning	LLMs can interpret quality/buffering patterns from radio-QoE sequences, e.g., explaining quality upgrades from RSRP/SNR/RSRQ/bitrate values.	Demonstrates LLM applicability to QoE estimation.
Network-to-Text Summarization	Radio metrics are pre-classified into 4 levels (Excellent/Good/Fair/Poor) to support natural-language summarization of sessions for operators.	Automatic experiment summaries (e.g., quality drops linked to poor RSRP/RSRQ).
LLM Agent for Monitoring	An LLM-based agent can observe QoE and radio metrics in real time and recommend corrective actions.	Intelligent monitoring with degradation alerts.

modeling, adaptive bitrate analysis, and learning-based prediction of user experience in next-generation wireless systems. Despite these contributions, the dataset's limitation is that it was collected from a single user device, with a single user streaming the sessions, which limits its generalizability across users, devices, mobility conditions, and viewing behaviors. As future work, the dataset can be extended to include mobility scenarios, greater content diversity, and larger-scale experiments to further enhance its value for benchmarking and model development.

REFERENCES

[1] M. Almquist *et al.*, "The prefetch aggressiveness tradeoff in 360 video streaming," in *Proceedings of the 9th ACM Multimedia Systems Conference*, pp. 258–269, 2018.

[2] Y. Bao *et al.*, "Shooting a moving target: Motion-prediction-based transmission for 360-degree videos," in *2016 IEEE International Conference on Big Data (Big Data)*, pp. 1161–1170, IEEE, 2016.

[3] X. Corbillon *et al.*, "Optimal set of 360-degree videos for viewport-adaptive streaming," in *Proceedings of the 25th ACM international conference on Multimedia*, pp. 943–951, 2017.

[4] X. Corbillon *et al.*, "Viewport-adaptive navigable 360-degree video delivery," in *2017 IEEE international conference on communications (ICC)*, pp. 1–7, IEEE, 2017.

[5] M. Graf *et al.*, "Towards bandwidth efficient adaptive streaming of omnidirectional video over http: Design, implementation, and evaluation," in *Proceedings of the 8th ACM on Multimedia Systems Conference*, pp. 261–271, 2017.

[6] S. Petrangeli *et al.*, "An http/2-based adaptive streaming framework for 360 virtual reality videos," in *Proceedings of the 25th ACM international conference on Multimedia*, pp. 306–314, 2017.

[7] F. Qian *et al.*, "Optimizing 360 video delivery over cellular networks," in *Proceedings of the 5th Workshop on All Things Cellular: Operations, Applications and Challenges*, pp. 1–6, 2016.

[8] F. Qian *et al.*, "Flare: Practical viewport-adaptive 360-degree video streaming for mobile devices," in *Proceedings of the 24th Annual International Conference on Mobile Computing and Networking*, pp. 99–114, 2018.

[9] L. Xie *et al.*, "360probdash: Improving QoE of 360 video streaming using tile-based http adaptive streaming," in *Proceedings of the 25th ACM international conference on Multimedia*, pp. 315–323, 2017.

[10] L. Xie *et al.*, "Cls: A cross-user learning based system for improving qoe in 360-degree video adaptive streaming," in *Proceedings of the 26th ACM international conference on Multimedia*, pp. 564–572, 2018.

[11] M. V. d. Broeck *et al.*, "It's all around you: Exploring 360 video viewing experiences on mobile devices," in *Proceedings of the 25th ACM international conference on Multimedia*, pp. 762–768, 2017.

[12] R. U. Mustafa *et al.*, "Investigating the impact of channel metrics in 5g nsa and sa on video streaming qoe," in *NOMS 2025-2025 IEEE Network Operations and Management Symposium*, pp. 1–7, IEEE, 2025.

[13] X. Corbillon *et al.*, "360-degree video head movement dataset," in *Proceedings of the 8th ACM on Multimedia Systems Conference*, pp. 199–204, 2017.

[14] W.-C. Lo *et al.*, "360 video viewing dataset in head-mounted virtual reality," in *Proceedings of the 8th ACM on Multimedia Systems Conference*, pp. 211–216, 2017.

[15] C. Wu *et al.*, "A dataset for exploring user behaviors in vr spherical video streaming," in *Proceedings of the 8th ACM on Multimedia Systems Conference*, pp. 193–198, 2017.

[16] E. J. David *et al.*, "A dataset of head and eye movements for 360 videos," in *Proceedings of the 9th ACM multimedia systems conference*, pp. 432–437, 2018.

[17] Y. Jin *et al.*, "Where are you looking? a large-scale dataset of head and gaze behavior for 360-degree videos and a pilot study," in *Proceedings of the 30th ACM International Conference on Multimedia*, pp. 1025–1034, 2022.

[18] I. Agtzidis *et al.*, "360-degree video gaze behaviour: A ground-truth data set and a classification algorithm for eye movements," in *Proceedings of the 27th ACM international conference on multimedia*, pp. 1007–1015, 2019.

[19] S. Fremerey *et al.*, "Avtrack360: An open dataset and software recording people's head rotations watching 360 videos on an hmd," in *Proceedings of the 9th ACM multimedia systems conference*, pp. 403–408, 2018.

[20] B. J. Li *et al.*, "A public database of immersive vr videos with corresponding ratings of arousal, valence, and correlations between head movements and self report measures," *Frontiers in psychology*, vol. 8, p. 2116, 2017.

[21] A. T. Nasrabadi *et al.*, "A taxonomy and dataset for 360 videos," in *Proceedings of the 10th ACM Multimedia Systems Conference*, pp. 273–278, 2019.

[22] W. Zhang *et al.*, "Leader360v: The large-scale, real-world 360 video dataset for multi-task learning in diverse environment," *arXiv preprint arXiv:2506.14271*, 2025.